Week 7 Lecture 14

Normal Distribution & Central Limit Theorem

Probability Theory and Mathematical Statistics

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> “Everyone believes in the Normal law of errors, the experimenters because they think it is a mathematical theorem, the mathematicians because they think it is an experimental fact.”

French mathematician Jules Henri Poincaré

Normal Distribution

$\mu =$ expectation, location parameter

$\sigma =$ standard deviation, scale parameter

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), -\infty < x < \infty$$

$E(X) = \mu$

$Var(X) = \sigma^2$

Standard Normal Distribution

Normal distribution with "standard parameters" $\mu = 0$ and $\sigma = 1$ is called **Standard Normal distribution**.

$Z =$ Standard Normal random variable

$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, Standard Normal pdf

$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \, dz$, Standard Normal cdf

Standard Normal Variable Transformation

A Standard Normal variable, usually denoted by Z, can be obtained from a non-standard Normal(μ, σ) random variable X by standardizing, that is, subtracting the mean and dividing by the standard deviation:

$$Z = \frac{X - \mu}{\sigma}.$$

Unstandardizing Z, we can reconstruct the initial variable X:

$$X = \mu + \sigma Z.$$

Using these transformations, any Normal random variable can be obtained from a Standard Normal variable Z.

Computing Standard Normal probabilities

For a Standard Normal random variable Z,

$$P(Z < 1.35) =$$

$$P(Z > 1.35) =$$

$$P(-0.77 < Z < 1.35) =$$

🖊 Computing non-standard Normal probabilities

Suppose that the average household income in some country is 900 coins, and the standard deviation is 200 coins. Assuming the Normal distribution of incomes, compute the proportion of “the middle class,” whose income is between 600 and 1200 coins.

🖊 Inverse problem

The government of the country from previous task decides to issue food stamps to the poorest 3% of households. Below what income will families receive food stamps?

Central Limit Theorem

Consider sums of random variables, denoted as $S_n = X_1 + \ldots + X_n$, appear in many applications.

Let $\mu = E(X_i)$ and $\sigma = \mathrm{Std}(X_i)$ for all $i = 1, \ldots, n$

How does S_n behave for large n?

Applet 🔗

Central Limit Theorem

Let $X_1, X_2, \ldots$ be independent random variables with the same expectation $\mu = E(X_i)$ and the same standard deviation $\sigma = \mathrm{Std}(X_i)$, and let

$$S_n = \sum_{i=1}^{n} X_i = X_1 + \ldots + X_n.$$

As $n \to \infty$, the standardized sum

$$Z_n = \frac{S_n - E(S_n)}{\mathrm{Std}(S_n)} = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution to a Standard Normal random variable, that is,

$$F_{Z_n}(z) = P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z\right)$$

for all z.

As long as n is large (the rule of thumb is $n > 30$), one can use Normal distribution to compute probabilities about S_n.

🖊 Allocation of disk space

A disk has free space of 330 megabytes.
Is it likely to be sufficient for 300 independent images, if each image has expected size of 1 megabyte with a standard deviation of 0.5 megabytes?

Elevator

You wait for an elevator, whose capacity is 907 kilograms.
The elevator comes with ten adult passengers. Suppose your own weight is 68 kilograms, and you heard that human weights are normally distributed with the mean of 75 kilograms and the standard deviation of 9 kilograms. Would you board this elevator or wait for the next one?

Normal Approximation to Binomial Distribution

Binomial variables represent a special case of $S_n = X_1 + \ldots + X_n$, where all X_i have Bernoulli distribution with some parameter p.

Small p: Approximate Binomial distribution with Poisson

Large p: Approximate Binomial distribution with Poisson using number of failures

For moderate values of p (e.g., $0.05 \leq p \leq 0.95$) and for large n, we can use:

$$\text{Binomial}(n, p) \approx \text{Normal}\left(\mu = np, \sigma = \sqrt{np(1-p)}\right)$$

Continuity Correction

Needed when approximating a discrete distribution (Binomial) by a continuous distribution (Normal).
Probability $P(X = x)$ may be positive if X is discrete, but it's always 0 for continuous X.
Direct use of the Normal approximation may lead to poor approximation.

Expand the interval by 0.5 units in each direction, then use the Normal approximation:

$$P_X(x) = P(X = x) = P(x - 0.5 < X < x + 0.5)$$

holds for a Binomial variable X.
Continuity correction doesn't change the event and preserves its probability.

🚀 New PC Virus

A new computer virus attacks a folder consisting of 200 files. Each file gets damaged with probability 0.2 independently of other files. What is the probability that fewer than 50 files get damaged?

Review

Reboot Time Distribution

The time, in minutes, it takes to reboot a certain system is a continuous variable with the density:

$$f(x) = \begin{cases} C(10-x)^2 & \text{if } 0 < x < 10 \\ 0 & \text{otherwise} \end{cases}$$

(a) Compute C.

(b) Compute the probability that it takes between 1 and 2 minutes to reboot.

Satellite

Consider a satellite whose work is based on a certain block A. This block has an independent backup B. The satellite performs its task until both A and B fail. The lifetimes of A and B are exponentially distributed with the mean lifetime of 10 years.

(a) What is the probability that the satellite will work for more than 10 years?

(b) Compute the expected lifetime of the satellite.

🚀 Lifetimes of computer memory chips

Lifetimes of computer memory chips have Gamma distribution with expectation $\mu = 12$ years and standard deviation $\sigma = 4$ years.

What is the probability that such a chip has a lifetime between 8 and 10 years?

Let Z be a Standard Normal random variable.

Compute

(a) $P(Z < 1.25)$

(b) $P(Z \leq 1.25)$

(c) $P(Z > 1.25)$

(d) $P(|Z| \leq 1.25)$

For a Normal random variable X with $E(X) = -3$ and $Var(X) = 4$, compute

(a) $P(X \leq 2.39)$

(b) $P(|X| \geq 2.39)$

(c) $P(|X + 3| \geq 2.39)$

Installation of some software package requires downloading 82 files. On the average, it takes 15 sec to download one file, with a variance of 16 sec^2. What is the probability that the software is installed in less than 20 minutes?

A computer lab has two printers. Printer I handles 40% of all the jobs. Its printing time is Exponential with the mean of 2 minutes. Printer II handles the remaining 60% of jobs. Its printing time is Uniform between 0 minutes and 5 minutes. A job was printed in less than 1 minute. What is the probability that it was printed by Printer I?